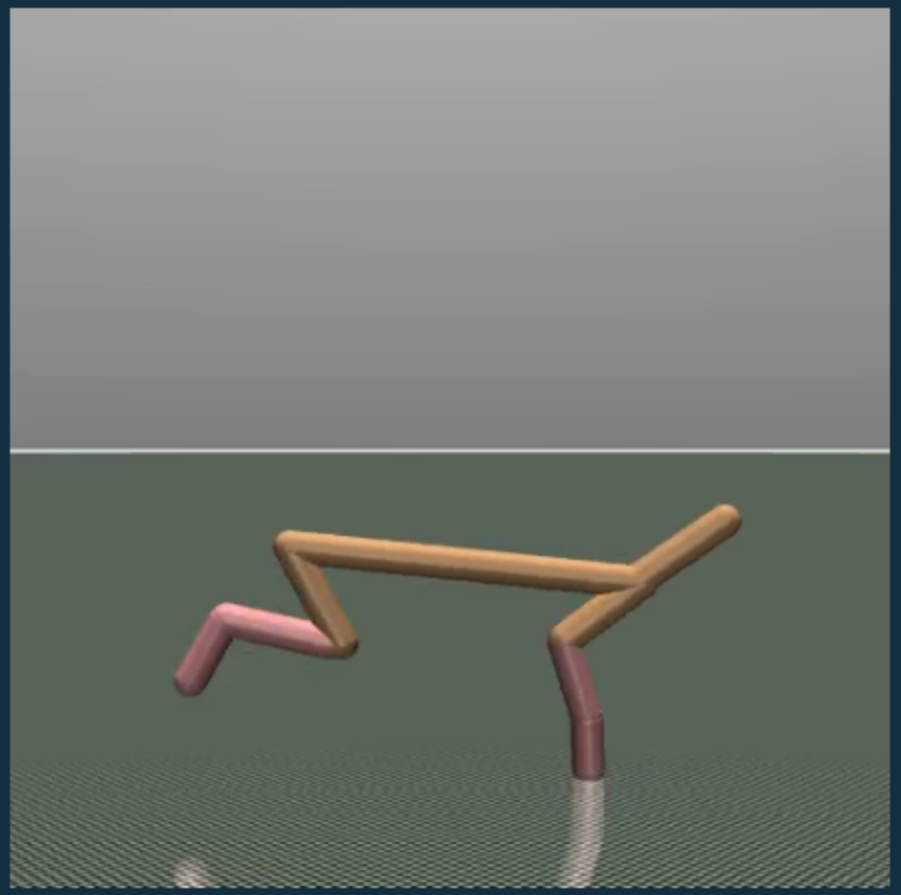
Implicit Q-Learning

HalfCheetah-v5

Offline reinforcement learning proposal



Aleksandr Medvedev Nikita Shankin Leo Vesin
Ruslan Nasibullin Telman Nuruzov

Offline data, learned values, continuous control

OFFLINE Model-free actor-critic

Recorded data

(s, a, r, next s)

→

IQL learner

Values + actor

→

Control policy

6 continuous torques

↓

Simulator evaluation

Full policy rollouts

Training: fixed transitions only

Better recorded actions receive more weight

Three updates inside IQL

01

Value V

Upper expectile

Favors stronger actions **0.7**

02

Twin critics

Q1 and Q2

reward + 0.99 × V(next state)

Bootstrap unless truly terminated

03

Actor

Weighted imitation

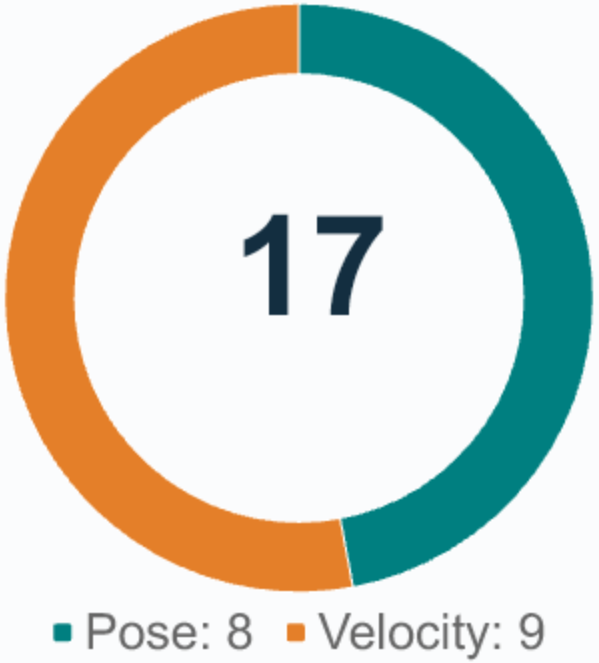
min(exp(3 × advantage), 100)

Advantage = min(target Q1, Q2) - V

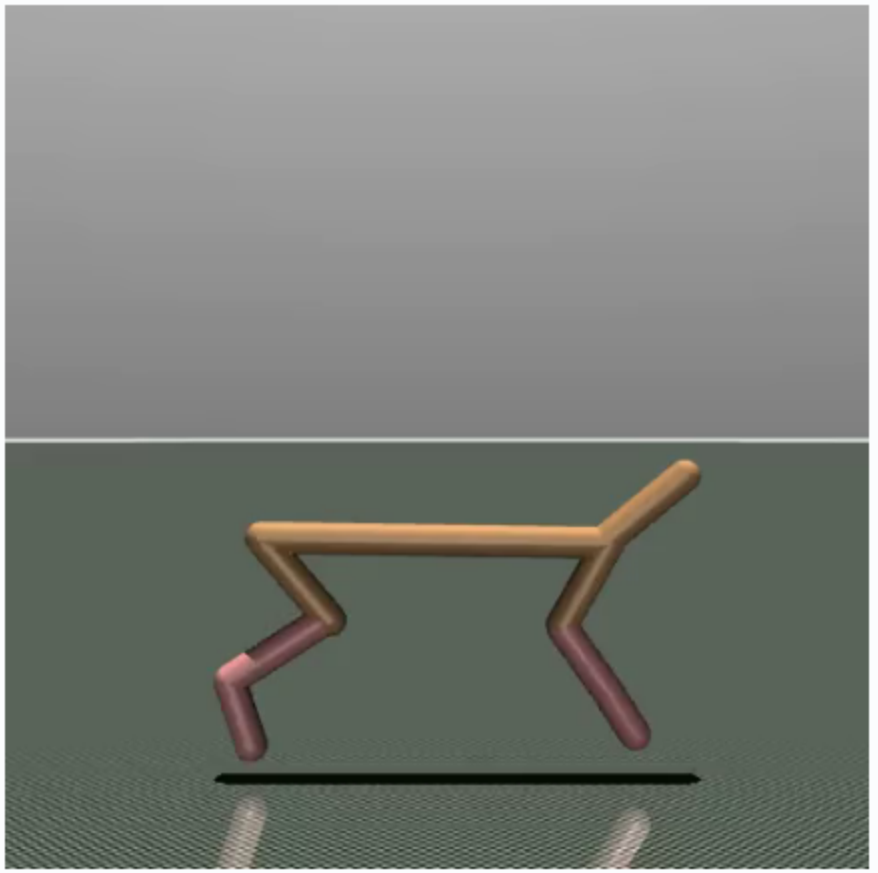
Policy output: tanh-squashed Gaussian

17 observations, 6 joint controls

Observation components



Absolute x excluded



2 legs × (thigh + shin + foot)

6

joint torques

[-1, 1]

per joint

Forward motion earns reward

Reward = forward velocity - 0.1 × sum(action²)

Illustrative step: six action components = 0.5

5.00	–	0.15	=	4.85
forward velocity		control cost		net reward

Goal: sustained, efficient forward motion

0

1,000 steps ≈ 50 seconds

A fall does not end the episode

A reproducible training and evaluation plan

1M
transitions

3
training seeds

500k
updates per seed

Minari medium dataset: 1,000 episodes. Batch size: 256.

Train
Seeds 0, 1, 2

→

Select
10 episodes / checkpoint

→

Final test
Separate episode seeds

Raw return + random baseline + full videos

Dataset-matched simulator: MuJoCo 3.2.3

Why this algorithm fits this task

Continuous torques → **Continuous actor**

Recorded locomotion → **Offline learning**

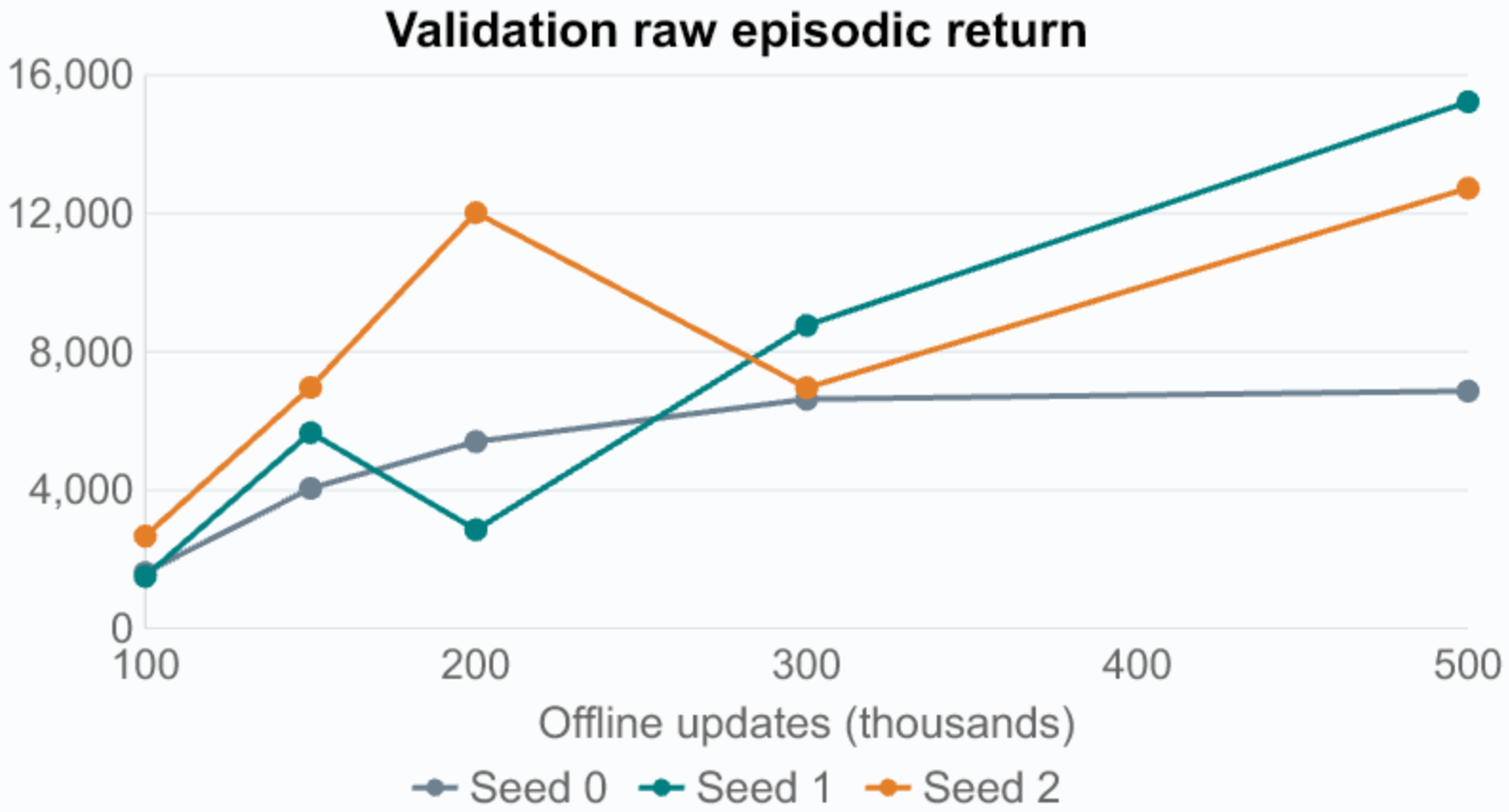
Returns + visible gait → **Measurable evaluation**

Challenges: data coverage, seed failures, physics differences

Multiple seeds, full episodes and pinned dependencies

Aleksandr: integration Nikita: data Leo: IQL
Ruslan: evaluation Telman: notebook and presentation

Learning progress and final test



15,260

final-test mean, 500k

vs 2,026 at 100k

10/10 → 0/10

episodes with long stationary tails

Means only: 10 episodes per point. Colab, MuJoCo 3.2.3.

Final test: seed 1, 10 new episodes. Other training seeds still fail.